Shri Vile Parle Kelavani Mandal's

# Institute of Technology, Dhule.

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## Department of Computer Engineering

**Academic Year: 2026-27**

**Programme Name- CSE-AIML**

**LABORATORY : Machine Learning Laboratory(26AF1245PCL508) SEMESTER : 5th Sem**

## List of Practical

| SR. NO. | PRACTICAL DESCRIPTION |
|---|---|
| 1 | Write a program to perform Exploratory Data Analysis, Data Visualization and Data Preprocessing through various feature-engineering algorithms in Python. |
| 2 | Implement simple and multiple linear regression using NumPy and using scikit-learn. Plot regression lines and residuals. Compute RMSE and $R^2$ values. |
| 3 | A. Implement a classification problem using decision boundaries (manual coding). Deduced accuracy, precision, recall, F1-score using Confusion Matrix.<br>B. Train a decision tree classifier using scikit-learn and visualize the tree. Analyze performance with/without pruning and depth control. Demonstrate overfitting using synthetic data. |
| 4 | Implement and visualize k-Nearest Neighbors classification. Tune k using k-fold cross-validation to a dataset and compare accuracy. Demonstrate curse of dimensionality. |
| 5 | Apply PCA and t-SNE for dimensionality reduction. Visualize high-dimensional data in 2D. Measure accuracy before and after feature reduction. |
| 6 | Implement a simple user-based and item-based collaborative filter. Use cosine similarity or Pearson correlation. Recommend top-N items for a user. |
| 7 | Train a linear and kernel (RBF, polynomial) SVM. Visualize margins and support vectors. Tune C and gamma hyperparameters. Analyze ROC curve and AUC. |
| 8 | A. Implement a single-layer perceptron from scratch. Train a multilayer perceptron using scikit-learn / Keras. Compare performance with different activation functions.<br>B. Build a 2–3 layer DNN for digit classification. Visualize layers, activations, and loss using Keras. |
| 9 | Implement and compare the Bagging (Random Forest) vs Boosting (AdaBoost/XGBoost) for a sample training dataset by performing hyper parameter tuning using GridSearch CV. Compute the accuracy of the classifier and Visualize model with Python. Visualize feature importance. |
| 10 | Perform K-Means with Elbow method, Agglomerative Clustering with plotting dendrograms. and visualize clusters. Find the silhouette score and DB Score to deduce internal and external quality of cluster. Fit a Gaussian Mixture Model (GMM) and visualize decision boundaries. |
| 11 | **Open Ended Assignment :** Create the Model Evaluation for a real life training data set stored as a .CSV file and by computing accuracy for different model proposed the best suitable machine learning algorithm for problem **or Solve the any real life problem suggested by Hackerearth site** |

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